Beyond Zoom: Cross-Tile Spatial-Relation Reasoning for Ultra-High-Resolution Remote-Sensing Vision-Language Models

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***Abstract*—Ultra-high-resolution (UHR) remote-sensing imagery provides detailed information about small objects and their spatial configuration, but its extreme image dimensions pose challenges for vision-language models (VLMs). Direct image resizing can remove fine-grained visual evidence, while independent tile-based processing can preserve local detail but disrupt spatial relationships between objects distributed across the scene. Existing UHR approaches primarily address evidence acquisition through resolution scaling, token selection, cropping, or active visual exploration. In contrast, this work investigates whether explicitly representing spatial relationships across independently processed image tiles can improve relational reasoning.**

We propose a lightweight cross-tile spatial-relation framework that combines question-conditioned region selection, local VLM reasoning, global coordinate transformation, and a geometric scene graph representing relationships between objects across tiles. Qwen2.5-VL-7B is used as the underlying VLM to provide a controlled baseline. The proposed method is evaluated on XLRS-Bench, with particular emphasis on object spatial relationship, relational counting, and multi-region grounding tasks. Controlled ablations compare direct inference, independent tiling, question-conditioned selection, and the proposed cross-tile relation representation under matched computational conditions.

[REPLACE: REPLACE WITH FINAL RESULTS: Briefly report the main quantitative improvement, task-specific improvement, and computational cost.]

The study aims to establish whether explicit geometric representation of cross-tile relationships provides a measurable and task-specific advantage for UHR remote-sensing VLM reasoning.

I. Introduction

Recent advances in vision-language models (VLMs) have enabled increasingly sophisticated visual understanding and reasoning across natural images and domain-specific applications. Remote sensing presents a particularly challenging setting because satellite imagery frequently contains large spatial extents, dense object distributions, and fine-grained structures that are difficult to preserve when images are processed at conventional resolutions.

Ultra-high-resolution (UHR) remote-sensing imagery makes this challenge more pronounced. XLRS-Bench introduced a dedicated benchmark for evaluating multimodal large language models on UHR remote-sensing scenes, with an average image size of approximately 8500×8500 pixels and 16 subtasks covering both perception and reasoning capabilities [1]. The benchmark demonstrates that current multimodal models still face substantial difficulties when required to interpret extremely large remote-sensing scenes.

A fundamental difficulty arises from the trade-off between local detail and global context. Processing an entire UHR image directly requires aggressive visual compression, which can remove information about small objects. Conversely, dividing the image into smaller tiles allows a model to inspect high-resolution regions but can disconnect objects that occur in different tiles. Consequently, a model may correctly identify individual objects while failing to determine their relationships at the scene level.

Recent research has primarily addressed the first part of this problem. GeoLLaVA-8K introduced background token pruning and anchored token selection to enable multimodal models to process remote-sensing imagery at resolutions up to $8K \times 8K$ [2]. GeoEyes further introduced on-demand visual focusing, allowing a model to actively zoom into task-relevant regions [3]. GeoVista extended active perception through global planning, branch-wise regional inspection, and explicit evidence tracking across multiple regions [4]. These approaches improve the acquisition and organization of visual evidence in UHR scenes.

However, evidence acquisition does not necessarily provide an explicit representation of geometric relationships between objects identified in independently processed regions. This distinction is important for tasks such as determining which object is north of another object, which facilities fall within a particular region, or how objects distributed across distant image tiles are spatially related.

Relational reasoning has previously been explored in remote-sensing visual question answering. EarthVQA introduced a relational reasoning framework that explicitly models object semantics and external spatial relationships using segmentation-derived representations and object-guided attention [5]. However, EarthVQA addresses conventional-resolution remote-sensing imagery rather than the cross-tile reasoning problem introduced by extremely large UHR scenes.

This work therefore investigates the following research question:

> *Does explicit cross-tile spatial-relation representation improve VLM performance on relational reasoning tasks in ultra-high-resolution remote-sensing imagery?*

We hypothesize that representing objects in a common global coordinate system and explicitly constructing their spa-

tial relationships will improve relational tasks while producing smaller gains on tasks that primarily require single-object perception or generic captioning.

The main contributions of this study are:

1) We formulate cross-tile spatial-relation reasoning as a specific challenge for UHR remote-sensing VLMs.
2) We introduce a lightweight geometric representation that connects objects extracted from independently processed tiles through a common global coordinate system.
3) We conduct controlled ablation and compute-matched experiments to isolate the contribution of explicit cross-tile spatial relationships.
4) We provide task-level and qualitative analysis to determine whether improvements are concentrated on relational reasoning tasks.

II. Related Work

A. Ultra-High-Resolution Remote-Sensing VLMs

The development of UHR remote-sensing benchmarks has highlighted limitations of existing multimodal models when processing very large satellite images. XLRS-Bench introduced manually annotated UHR imagery with an average image size of approximately 8500×8500 pixels and defined 16 subtasks covering perception and reasoning capabilities [1]. The benchmark provides a direct evaluation setting for studying the effect of image resolution and scene complexity on remote-sensing VLMs.

GeoLLaVA-8K addressed the computational challenges associated with UHR imagery through Background Token Pruning and Anchored Token Selection. The method exploits the observation that important visual information is often concentrated in object-centric regions and enables processing at resolutions up to $8K \times 8K$ [2]. These approaches primarily address the problem of preserving or efficiently accessing useful visual information under extreme image dimensions.

B. Active Visual Exploration for UHR Remote Sensing

A second line of research treats UHR interpretation as an active perception problem. GeoEyes introduced on-demand visual focusing using a staged training strategy and reinforcement learning to improve adaptive zooming and evidence acquisition [3]. GeoVista subsequently introduced planning-driven active perception, branch-wise regional inspection, and explicit evidence tracking to reduce incomplete exploration and duplicated evidence [4].

These methods demonstrate the importance of selecting and inspecting relevant regions rather than treating the complete UHR image uniformly. Nevertheless, their primary objective is evidence acquisition and tracking. The present work focuses on a complementary question: after evidence has been acquired from multiple regions, how should the spatial relationships among objects distributed across those regions be represented for final reasoning?

TABLE I
Task categories considered in the evaluation.

Category	Representative Tasks	Expected Effect
Relational	OSR, relational counting	High
Multi-region	Grounding, regional reasoning	High
Perception	Object recognition, properties	Low/Moderate
Captioning	Scene description	Low

C. Relational Reasoning in Remote Sensing

Relational reasoning has also been investigated in remote-sensing VQA. EarthVQA introduced a multi-task dataset containing 6,000 images and 208,593 question-answer pairs and proposed a Semantic Object Awareness framework for object-centric relational reasoning [5]. The framework uses segmentation-derived object representations and cross-attention to model object relationships.

While EarthVQA establishes the importance of explicit object relationships for remote-sensing reasoning, it does not specifically address the spatial discontinuity created when an extremely large image is divided into independently processed tiles. Our work investigates this intersection between UHR visual processing and explicit geometric relational reasoning.

III. Dataset

A. XLRS-Bench

XLRS-Bench is used as the primary evaluation benchmark [1]. The benchmark contains extremely large remote-sensing images with an average spatial dimension of approximately 8500×8500 pixels and evaluates multimodal models across 16 subtasks.

The benchmark covers both perceptual and reasoning capabilities. For this study, particular attention is given to tasks that require relationships between multiple objects or spatial regions, including Object Spatial Relationship, relational and regional counting, and multi-region grounding.

Non-relational tasks such as object perception and captioning are also retained for comparison. This allows the study to determine whether the proposed method produces task-specific improvements rather than uniformly improving all types of visual understanding.

IV. Data Preprocessing

A. Image Standardization

Each XLRS-Bench image is retained at its native resolution for the proposed high-resolution processing pipeline. A downsampled representation is additionally generated to provide global scene context.

[REPLACE: REPLACE WITH YOUR EXACT PREPROCESSING SETTINGS: image format, native resolution statistics, normalization, resizing method, and hardware constraints.]

B. Image Tiling

Each UHR image is divided into overlapping high-resolution tiles. Tile size and overlap are treated as experimental parameters because they determine the trade-off between local visual detail, spatial coverage, computational cost, and duplicate object observations.

[REPLACE: REPLACE: TILE SIZE = XXX pixels; OVERLAP = XX%; NUMBER OF TILES = XXX.]

The spatial coordinates of each tile are retained so that object observations extracted from different tiles can subsequently be mapped to the coordinate system of the original image.

C. Question-Conditioned Region Selection

Rather than processing all tiles equally, candidate tiles are ranked according to their relevance to the input question. Relevant objects and regions identified from the global scene representation are used to prioritize high-resolution crops.

[REPLACE: REPLACE WITH THE FINAL TILE SELECTION METHOD AND SCORING FUNCTION.]

V. Proposed Method

A. Overview

The proposed framework consists of four principal stages: global scene understanding, question-conditioned tile selection, local visual reasoning, and cross-tile spatial-relation reasoning.

Given an input image I and question q, the system first generates a low-resolution global representation:

$$G = f_{\text{global}}(I). \tag{1}$$

The global representation provides scene-level context and supports selection of high-resolution regions relevant to the question.

A set of candidate tiles is then generated:

$$T = \{t_1, t_2, \ldots, t_N\}. \tag{2}$$

A question-conditioned selection function ranks the candidate regions:

$$S_i = f_{\text{select}}(t_i, q, G), \tag{3}$$

and the top-K regions are selected for detailed processing.

B. Local Visual Reasoning

Each selected high-resolution tile is processed independently using Qwen2.5-VL-7B [6]. For each tile, the model is prompted to identify relevant objects, attributes, and spatial evidence.

For an object o_i detected in tile t_j, its local coordinates are transformed into global image coordinates using the known tile offset:

$$B_i^{global} = B_i^{local} + (x_j, y_j), \tag{4}$$

where (x_j, y_j) denotes the top-left coordinate of tile t_j in the original image.

C. Cross-Tile Spatial Relation Graph

The central component of the proposed framework is a geometric spatial relation graph.

Let

$$\mathcal{G} = (V, E) \tag{5}$$

represent the scene graph, where each node $v_i \in V$ represents an object and each edge $e_{ij} \in E$ represents a spatial relationship between two objects.

Each object is represented by its identity, attributes, global bounding box, and center coordinate:

$$v_i = (c_i, a_i, B_i, p_i). \tag{6}$$

Spatial relationships are computed geometrically using the global coordinates. Depending on the task, the graph can encode relationships such as direction, distance, adjacency, containment, and intersection.

For example,

$$e_{ij} = r(o_i, o_j), \tag{7}$$

where r denotes a spatial relation between objects o_i and o_j.

Because all objects are represented in a common coordinate system, objects identified in different tiles can be directly compared.

D. Global-Local Relation Fusion

The final reasoning stage receives three forms of evidence:

1) global scene context,
2) selected high-resolution visual evidence,
3) the cross-tile spatial relation graph.

The final response is generated as

$$y = f_{\text{VLM}}(G, T_{selected}, \mathcal{G}, q). \tag{8}$$

This design separates visual evidence acquisition from explicit spatial relation construction and allows the contribution of the relation representation to be evaluated independently.

VI. Experimental Design

Four configurations are evaluated.

A. Direct VLM

The original image representation is provided directly to Qwen2.5-VL-7B without tile-level processing.

B. Independent Tiling

The image is divided into high-resolution tiles and processed independently. No explicit cross-tile spatial representation is provided.

C. Question-Conditioned Tiling

Question-conditioned region selection is introduced before local VLM reasoning, but no explicit spatial relation graph is constructed.

TABLE II
EXPERIMENTAL CONFIGURATIONS.

Method	Direct	Tiling	Selection	Relations
Direct VLM	✓			
Independent Tiling		✓		
Question-Conditioned Tiling		✓	✓	
Proposed Method		✓	✓	✓

TABLE III
OVERALL PERFORMANCE ON XLRS-BENCH.

Method	Overall Score	Relational Score
Direct VLM	[REPLACE: XX.XX]	[REPLACE: XX.XX]
Independent Tiling	[REPLACE: XX.XX]	[REPLACE: XX.XX]
Question-Conditioned Tiling	[REPLACE: XX.XX]	[REPLACE: XX.XX]
Proposed Method	[REPLACE: XX.XX]	[REPLACE: XX.XX]

D. Proposed Method

The complete framework combines question-conditioned tile selection, local VLM reasoning, global coordinate transformation, and cross-tile spatial relation representation.

VII. Ablation Studies

The primary ablation removes the cross-tile relation representation while keeping the VLM, tile selection mechanism, image regions, and computational budget unchanged.

Additional experiments investigate the effects of:

- tile size,
- tile overlap,
- number of selected tiles,
- object localization accuracy,
- relation types,
- computational budget.

The central hypothesis predicts that removing the relation representation will primarily affect tasks requiring relationships among spatially separated objects.

VIII. Computational Evaluation

Because the proposed framework may involve multiple VLM calls, performance will be evaluated alongside computational cost.

The following quantities will be reported:

- task accuracy,
- number of VLM calls,
- inference time,
- GPU memory consumption,
- computational budget.

Where possible, comparisons will be performed under matched computational budgets to determine whether improvements arise from the proposed spatial reasoning mechanism rather than simply from processing more visual regions.

[REPLACE: REPLACE WITH FINAL HARDWARE, INFERENCE TIME, MEMORY, AND COMPUTE RESULTS.]

IX. Results

A. Overall Performance

[REPLACE: REPLACE WITH FINAL XLRS-BENCH OVERALL RESULTS AND COMPARISON WITH BASELINES.]

B. Task-Level Performance

[REPLACE: REPLACE WITH TASK-SPECIFIC RESULTS FOR OBJECT SPATIAL RELATIONSHIP, COUNTING, GROUNDING, CAPTIONING, AND OTHER RELEVANT TASKS.]

C. Ablation Results

[REPLACE: REPLACE WITH ABLATION RESULTS SHOWING THE EFFECT OF REMOVING THE CROSS-TILE RELATION REPRESENTATION.]

D. Computational Results

[REPLACE: REPLACE WITH COMPUTE-MATCHED RESULTS, INFERENCE TIME, VLM CALLS, AND MEMORY REQUIREMENTS.]

X. Error Analysis

The proposed framework will be evaluated through qualitative and quantitative analysis of failure cases. Errors will be categorized into fine-grained detail loss, tile-boundary failures, incorrect spatial relations, duplicate object observations, and errors propagated from object localization.

[REPLACE: REPLACE WITH FINAL ERROR ANALYSIS AND REPRESENTATIVE FAILURE CASES.]

XI. Application Demonstration

To demonstrate practical relevance, the proposed framework will be evaluated in a flood-impact scenario involving infrastructure distributed across an UHR scene.

The system will be queried with questions such as which schools or health facilities are located within an inundated region. Such queries require combining infrastructure objects and hazard regions that may be observed in different image tiles.

[REPLACE: REPLACE WITH FLOOD CASE STUDY LOCATION, DATA SOURCE, IMAGE DETAILS, QUESTION SET, AND QUALITATIVE RESULTS.]

XII. Discussion

The central objective of this study is not to develop a new foundation VLM, but to isolate the contribution of explicit spatial relation representation in UHR remote-sensing reasoning.

If the proposed method produces its largest improvements on relational tasks while providing smaller improvements on non-relational tasks, this would support the hypothesis that cross-tile relation loss represents a distinct limitation of tile-based UHR reasoning.

Conversely, if improvements are observed uniformly across all tasks, the results would suggest that the observed benefit may arise from more general improvements in visual evidence acquisition rather than from the proposed relational mechanism.

[REPLACE: REPLACE WITH FINAL INTERPRETATION OF RESULTS AND COMPARISON WITH THE HYPOTHESIS.]

XIII. Conclusion

This study proposes a cross-tile spatial-relation reasoning framework for ultra-high-resolution remote-sensing vision-language understanding. The framework combines question-conditioned region selection, local high-resolution visual reasoning, global coordinate transformation, and geometric relation construction.

The research is designed to determine whether explicitly preserving spatial relationships between objects across independently processed image tiles can improve relational reasoning in UHR remote-sensing scenes.

[REPLACE: REPLACE WITH FINAL CONCLUSION BASED ON EXPERIMENTAL RESULTS.]

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